MILITARY SPECIFICATION

TEST SET, TORPEDO, MARK 281 MOD 1

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers the Torpedo Test Set Mk 281 Mod 1, which provides for presetting of all synchronous and nonsynchronous electrical functions of torpedoes. (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing; Aeronautical and Associated Equipment, General Specification for

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-761	Electrical Power, Alternating Current for Shipboard Use, Characteristics and Utilization of
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment

DRAWINGS

Naval Ordnance Systems Command (Code Ident 10001)

2639900	Mark 281 Mod 1, Test Set and Case

PUBLICATIONS

Naval Ordnance Systems Command (Code Ident 10001)

OD 9154	Specifications for Electrical Setting of Torpedoes

(Copies of specifications, standards, drawings, and publications required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The Torpedo Test Set Mk 281 Mod 1, hereinafter referred to as the "test set," is composed of six major subassemblies as defined on Drawing 2639900. During torpedo testing it is used to:

(a) Set and check the synchronous functions of enable range, gyro angle, and depth order.

(b) Set and check the nonsynchronous functions of horizontal search, speed, homing, stratum limit, and anticapture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

(c) Apply firing voltage.

(d) Convert alternating-current (AC) power to direct-current (DC) power.

(e) Provide torpedo warmup power.

3.2 First article requirement. Unless otherwise specified in the contract or purchase order, three first article samples of the test set shall be manufactured using the methods and procedures proposed for production. These samples will be tested as specified in section 4 to determine if the proposed production method can yield an item which meets the technical and design requirements of this specification and the contract or purchase order.

3.3 General requirements.

3.3.1 Materials and components. All materials and purchased parts entering into the final assembly of the test set shall conform to the detailed requirements of Drawing 2639900.

3.3.2 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions for the test set shall be in accordance with Drawing 2639900 and MIL-STD-454, requirements 1, 3, 15, and 62, as applicable.

3.3.3 Moisture, fungus, and corrosion resistance. Moisture, fungus, and corrosion resistance shall be in accordance with detail requirements of Drawing 2639900 and MIL-STD-454, requirements 4, 16, and 31, as applicable.

3.3.4 Marking. Marking of assemblies, subassemblies, and parts shall be as specified by Drawing 2639900 and MIL-STD-454, requirement 34, as applicable.

3.3.5 Interchangeability. Interchangeability shall be as specified in MIL-STD-454, requirement 7. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings shall be made without prior revisions and correction of the applicable drawings. (See 3.3.7.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

3.3.6 Gages. The contractor shall provide himself with all necessary and adequate gages to ensure that the material defined in this specification will meet the dimensional requirements shown on the applicable drawings.

3.3.7 Conflicting requirements. Conflicting requirements arising between this specification and any publications or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, correction, or resolution.

3.4 Performance requirements.

3.4.1 Input power. The test set shall meet all performance requirements with the POWER INPUT receptacle supplied with the following power, as defined in MIL-STD-761:

(a) Type I, 1ϕ, 60 Hertz (Hz), 115 volts alternating current (VAC), through an isolation transformer, to pins A and B

(b) Type III, 3ϕ, 400 Hz, 115 VAC, phase order ABC, with phase A to pin C, phase B to pin D, and phase C to pin E

(c) Type I, 1ϕ, 60 Hz, 115 VAC, to pins G and H.

3.4.1.1 The test set shall meet all performance requirements without drawing more than the following amperages at the POWER INPUT receptacle:

(a) Pins A and B: 15 amps
(b) Pins C, D, and E: 7 amps
(c) Pins G and H: 20 amps.

3.4.1.2 See 4.8.1 for test procedure.

3.4.2 Synchronous servomechanism requirements.

3.4.2.1 Gyro angle order. The test set shall be capable of setting single-speed and double-speed gyro systems. In setting the systems, the accuracy and synchronizing time shall be as specified in OD 9154, First Revision. The limiting circuit shall prevent the setting of a gyro order greater than 160.00° ± 0.01° in either direction when operating with either system. An indicator lamp shall light when the limit circuit has inhibited setting capability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

3.4.2.2 Depth order. The test set shall be capable of setting single-speed low and single-speed high depth systems. In setting single-speed low systems, the accuracy and synchronizing time shall be as specified in OD 9154, First Revision. In setting single-speed high systems, the static accuracy shall be 3.0 feet and the synchronizing time for a 40-foot order shall be no more than 20 seconds. The limiting circuit shall cause the OFF lamp to come on for any order less than 0 or greater than 999.9 feet for all depth systems, unless a ground is present on pin z, t, or AL of the OUTPUT receptacle, in which case the corresponding limits shall be 0 and 1999.9 feet. An indicator lamp shall light when the limit circuit has inhibited setting capability.

3.4.2.3 Enable run order. The test set shall be capable of setting enable run systems. In setting the systems, the accuracy and synchronizing time shall be as specified in OD 9154, First Revision.

3.4.2.4 Order indicators. The test set shall indicate the positions to which the gyro angle, depth, and enable run orders have been set. When operating in the SCAN mode, the indicators, in conjunction with the ORDER ERROR meter, shall indicate the true positions within the following mean errors: (1) two-speed gyro, ±0.01°; (2) single-speed gyro, ±0.10°; (3) high-speed depth, ±0.10 foot; (4) low-speed depth, ±1.0 foot; and (5) enable run, ±10 yards.

3.4.2.5 Synchronous SCAN/SET switch. The synchronous SCAN/SET switch shall allow scanning of the synchronous function settings to check the setting without having the torpedo synchros follow in correspondence. The SCAN/SET switch shall have an electrical interlock that prevents the switch from operating in the SET position when power is initially applied and when switching to or from the enable function.

3.4.3 Nonsynchronous presetting requirements.

3.4.3.1 Three-position steppers. The test set shall be capable of providing 24 ± 2.4 volts direct current (VDC) and 2.40 amps (max) to each of the selection circuits to accomplish presetting of homing, stratum, speed and horizontal search modes in the applicable torpedoes. The unit shall visually indicate the following individual settings of each function:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

Homing	Stratum
Listen	No limit
Echo	Torpedo above limit
Combination	Torpedo below limit

Speed	Horizontal search
Low	Circle port
Medium	Circle starboard
High	Straight

3.4.3.2 Eight-position stepper. The test set shall be capable of providing 24 ± 2.4 VDC and 3.60 amps (max) to the anticapture selection circuit to accomplish presetting the seven anticapture modes of Torpedo Mk 37 Mods 2 and 3. The test set unit shall visually indicate each of the seven settings.

3.4.3.3 Nonsynchronous SCAN/SET switch. The nonsynchronous SCAN/SET switch shall allow scanning of the nonsynchronous functions to locate initial preset positions without activating torpedo stepping relays. The SCAN/SET switch shall have an electrical interlock that prevents the switch from operating in the SET position when power is initially applied.

3.4.4 Firing circuit. The test set shall be capable of providing from 106 to 121 VAC at a current of 6 ± 1 amps to the firing circuit. The FIRE/READY indicator shall indicate prefiring circuit continuity. When the firing switch is closed, torpedo service power shall be terminated.

3.4.4.1 The fire indicator lamp shall light if the firing switch is disabled in a manner that would apply firing voltage to the attached torpedo if the OPERATION FUNCTION switch were turned to the TORP PWR ON position.

3.4.5 Tube solenoid circuit. When the firing circuit is closed, the test set shall be capable of providing from 106 to 121 VAC at a current of 17 ± 2 amps to energize the tube solenoid.

3.4.6 Battery heater circuit. The test set shall be capable of providing 115 ± 6 VAC at a maximum current of 3 amps to energize the torpedo battery heater. An indicator lamp shall indicate voltage presence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

3.4.7 24-VDC power. The test set shall provide from 18 to 32 VDC at a maximum current of 10 amps.

3.4.8 400-Hz protection circuit. The test set shall provide a visual indication of 400-Hz power status and shall disconnect 400-Hz power from the torpedo if any of the following conditions exist: phase rotation is not ABC, any phase is missing, line-to-line voltage is greater than 135 volts, or line-to-line voltage is less than 95 volts.

3.5 Size. The test set, with its cover removed, shall pass through a circular hole with a diameter of 25 ± 1/32 inches.

3.6 Weight. The complete test set, including case, cable, and spares shall weigh no more than 220 pounds.

3.7 Life. The test set should have a minimum storage life of 5 years. During this storage, the test set may be exposed to any or all of the environments of 3.8.

3.8 Environmental requirements.

3.8.1 Vibration. The test set shall show no damage and meet all performance and product requirements after the vibration test of 4.10.1.

3.8.2 Shock. The test set shall show no damage and meet all performance and product requirements after the shock test of 4.10.2.

3.8.3 Temperature and humidity. The test set shall show no damage and shall meet all performance and product requirements after the exposure tests of 4.10.3 and 4.10.4.

3.8.4 Salt-fog. The test set shall show no evidence of corrosion and shall meet all performance and product requirements after the exposure test of 4.10.5.

3.9 Workmanship. The test set, including parts and accessories, shall be manufactured and finished in a workmanlike manner in accordance with MIL-STD-454, requirement 9. All material shall be of sound and uniform quality and condition and shall contain no defects which can affect performance or serviceability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. Inspection and testing of the test set shall be classified as follows:

(a) First article inspection
(b) Quality conformance inspection.

4.2.1 First article inspection. Unless otherwise specified in the contract or purchase order, a representative sample of three test sets shall be manufactured using the methods and procedures proposed for production. These test sets shall be subjected to the first article inspection procedure of 4.5 and shall meet the requirement of 3.2.

4.2.2 Quality conformance inspection. The procedures of 4.6 shall be performed on each test set submitted for acceptance under contract (100% inspection) to assure that production units meet the requirements of this specification and the contract or purchase order.

4.3 Test equipment. The following items of test equipment are required to perform the inspections set forth herein:

(a) Multimeter or meters, isolated inputs, high impedance (10 megohms minimum), capable of measuring the following with 3% or better accuracy:

(1) 0.1 to 32 VDC
(2) 0.1 to 130 VAC root mean square (rms)
(3) 0.5 ohm or less
(4) 1 megohm or more

(b) Ammeter, AC, 5% or better accuracy, 25-amp full-scale (3 needed)

(c) Ammeter, AC, 5% or better accuracy, 10-amp full-scale (4 needed)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

(d) Resistor, 960-ohm ± 5%, 25-watt

(e) Resistor, 2.5-ohm ± 5%, 50-watt

(f) Rheostat, 250-ohm, 15-amp

(g) Resistor, decade, General Radio 1433-B, or equal: 1% resistors (see 4.8.15.12)

(h) Inductor, decade, General Radio 1490-F, or equal: 225 mH ± 10%, 50 ma, 20 ohms DC (max), Q = 20 (min) at 400 Hz (see 4.8.18.1)

(i) Ratio transformer, Gersch SS-2, or equal

(j) Power supply, adjustable, 400 Hz, 3ϕ delta, 2.5 kilovolt-ampere (KVA), 75 to 155 VAC (rms)

(k) Power supply, adjustable, current-limited, 24 ± 1 VDC, 0.5 to 2 amps.

4.4 Test conditions. Unless otherwise specified, the test set shall be inspected under the following ambient conditions:

(a) Temperature: 65° to 95° Fahrenheit
(b) Altitude: 0 to 5000 feet above sea level
(c) Humidity: 0% to 95% relative.

4.5 First article inspection. The inspections and tests required for first article inspection are indicated in table I, in the order to be performed.

4.5.1 Failures. A failure of any test set in any test or inspection shall be cause for denial of production approval.

4.6 Quality conformance inspection. The inspections and tests required for quality conformance inspections are indicated in table I, in the order to be performed.

4.6.1 Failures. A failure of any test set in any test or inspection shall be cause for rejection of that test set. Should more than two consecutive sets fail for the same defect, production shall be suspended until corrective measures are taken.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:41Z
Check the source to verify that this is the current version before use.

Table I

INSPECTION PROCEDURES

Test	Requirements	Method	First article	Quality conformance
Visual inspection	3.3, 3.5, 3.6, 3.9	4.7	X	X
Performance	3.4	4.8	X	
Accuracy	OD 9154, 3.4.2.4	4.9	X	
Vibration, shipping	3.8.1	4.10.1.1	X	
Shock	3.8.2	4.10.2	X	
Accuracy	OD 9154, 3.4.2.4	4.9	X	
Vibration, shipboard vertical axis	3.8.1	4.10.1.2	X	
Performance	3.4	4.8	X	
Vibration, shipboard depth axis	3.8.1	4.10.1.2	X	
Performance	3.4	4.8	X	
Vibration, shipboard width axis	3.8.1	4.10.1.2	X	
Performance	3.4	4.8	X	
Temperature, low	3.8.3	4.10.3.1	X	X
Performance	3.4	4.8	X	X
Temperature, high	3.8.3	4.10.3.2	X	X
Performance[1]	3.4	4.8	X	
Humidity, low	3.8.3	4.10.4.2	X	
Salt-fog	3.8.4	4.10.5	X	
Humidity, high[1]	3.8.3	4.10.4.3	X	
Accuracy	OD 9154, 3.4.2.4	4.9	X	X
Performance	3.4	4.8	X	X

[1]To be added to the procedure only if specifically called for in the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

4.7 Visual inspection procedure.

4.7.1 The complete test set, with case, cable, and spares, shall be weighed on a calibrated scale and shall conform to the requirement of 3.6.

4.7.2 With cover removed, the test set shall conform to the requirement of 3.5.

4.7.3 The test set shall be examined visually and shall conform to the requirements of 3.3.1, 3.3.2, 3.3.3, 3.3.4, and 3.9. If necessary to complete the examination, access doors or screwed-on plates shall be opened or removed.

4.7.3.1 Any evidence of material which has been treated to conceal defects shall be cause for rejection of the test set.

4.7.4 Using suitable gages, the test set shall be measured and shall conform to the requirements of 3.3.2.

4.8 Performance inspection procedure.

4.8.1 Input power test.

4.8.1.1 The following five meters shall be connected between the power supplies and the indicated pins of the POWER INPUT receptacle:

(a) Pins A and G: 25-amp (full-scale) AC ammeters
(b) Pins C, D, and E: 10-amp (full-scale) AC ammeters.

4.8.1.2 These five ammeters shall be monitored throughout the following test procedures, with special attention being given during the tests requiring significant power from the test set. At no time shall the current through the indicated POWER INPUT receptacle pin exceed the following:

(a) Pin A: 15 amps
(b) Pins C, D, or E: 7 amps each
(c) Pin G: 20 amps.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

4.8.2 Preoperation checkout.

4.8.2.1 Perform the following:

(a) Turn OPERATION FUNCTION switch to TORP PWR OFF.

(b) Remove any jumpers or other devices from the OUTPUT receptacle (hereinafter referred to as "OUTPUT").

(c) Set all test set circuit breakers ON.

(d) Turn all power supplies ON.

4.8.2.2 The following shall be noted:

(a) TORP PWR OFF lamp: on

(b) Cooling fan: running

(c) HOURS time indicator flywheel (at left end): turning

(d) OUTPUT: 0.1 VDC (max) from AC(+) to AD; 1 VAC (max) between AA and AB, AA and F, and AB and F.

4.8.2.3 Set the AC POWER circuit breaker OFF.

4.8.2.4 The following shall be noted:

(a) TORP PWR OFF lamp: off
(b) Cooling fan: stopping or stopped.

4.8.2.5 Return the AC POWER circuit breaker ON.

4.8.3 Synchronous Indicator Test.

4.8.3.1 Perform the following:

(a) Turn the OPERATION FUNCTION switch to SELF TEST.
(b) Press SYNC CONTROL OFF button.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.8.3.2 The following shall be noted:

(a) SELF TEST lamp: on
(b) BATTERY HEATER lamp: on
(c) SELF TEST 1 or SELF TEST 2 lamp: on (not both)
(d) SYNC OFF lamp: on
(e) SYNC SCAN lamp: on.

4.8.3.3 Press the SELF TEST 1/SELF TEST 2 button. This action shall reverse the SELF TEST mode indication existing at 4.8.3.2(c).

4.8.3.4 Press ENABLE RUN ORDER button. The SYNC OFF lamp shall go off and the ENABLE RUN ORDER lamp shall come on.

4.8.3.5 Press DEPTH SET ORDER button. The ENABLE RUN ORDER lamp shall go off and the DEPTH SET ORDER lamp shall come on.

4.8.3.6 Press GYRO ANGLE ORDER button. The DEPTH SET ORDER lamp must go off and the GYRO ANGLE ORDER lamp come on.

4.8.3.7 The following shall be noted:

(a) OUTPUT: 0.1 VDC (max) from AC(+) to AD; 1 VAC (max) between AA and AB, AA and F, and AB and F.

4.8.3.8 Turn the OPERATION FUNCTION switch to TORP PWR OFF.

4.8.4 Power on test.

4.8.4.1 Turn the OPERATION FUNCTION switch to TORP PWR ON. The following shall be noted:

(a) TORP PWR ON lamp: on

(b) BATTERY HEATER lamp: on

(c) ABC PHASE lamp: on

(d) READY lamp: on

(e) OUTPUT: 27 VDC (min) to 32 VDC (max) from AC(+) to AD; 115 ± 10 VAC between AB and AA, AB and F, and AA and F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.8.4.2 Set DC POWER circuit breaker OFF. The OUTPUT shall read 0.1 VDC (max) from AC(+) to AD.

4.8.4.3 Set DC POWER circuit breaker back ON, and turn the OPERATION FUNCTION switch to TORP PWR OFF.

4.8.5 Battery heater test.

4.8.5.1 Turn the OPERATION FUNCTION switch to BATTERY HEATER. The following shall be noted:

(a) BATTERY HEATER lamp: on

(b) TORP PWR OFF lamp: on

(c) OUTPUT: 0.1 VDC (max) from AC(+) to AD; 1 VAC (max) between AA and AB, AA and F, and AB and F.

4.8.5.2 Rotate the OPERATION FUNCTION switch through all its positions. The following shall be noted:

(a) BATTERY HEATER lamp: on, in all positions except TORP PWR OFF

(b) OUTPUT: 115 ± 10 VAC between AF and AI in all positions except TORP PWR OFF; 0.1 VAC (max) between AF and AI in TORP PWR OFF position.

4.8.5.3 Turn the OPERATION FUNCTION switch to TORP PWR OFF.

4.8.6 Monitor ready test.

4.8.6.1 Perform the following:

(a) Connect a jumper from AC to a on the OUTPUT.
(b) Turn the OPERATION FUNCTION switch to TORP PWR ON.

4.8.6.2 The MONITOR READY lamp shall come on.

4.8.6.3 Turn the OPERATION FUNCTION switch to TORP PWR OFF. Remove OUTPUT jumper.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.8.7 Limit test.

4.8.7.1 Perform the following:

(a) Connect a jumper from AC to AL on the OUTPUT.
(b) Turn the OPERATION FUNCTION switch to TORP PWR ON.

4.8.7.2 The following shall be noted:

(a) SPEED lamp: on
(b) MEDIUM lamp: on
(c) NONSYNC CONTROL lamp: on
(d) NONSYNC SCAN lamp: on
(e) SYNC SCAN lamp: on.

4.8.7.3 Press ENABLE RUN ORDER button and rotate ORDER ADJUST controls from one stop to the other. The digital indicators shall count in numeric sequence except for the 35999-00000 transition, and the OFF lamp shall not come on at any point.

4.8.7.4 Press the DEPTH SET ORDER button and rotate the ORDER ADJUST controls from one stop to the other. The digital indicators shall count in numeric sequence except for the 35999-00000 transition, and the OFF lamp shall be on for all numbers except those between 0000.0 and 1999.9, inclusive.

4.8.7.5 Press the GYRO ANGLE ORDER button and rotate the ORDER ADJUST controls from one stop to the other. The digital indicators shall count in numeric sequence except for the 35999-00000 transition, and the OFF lamp shall be off for all numbers except those between 160.00 and 199.99, inclusive.

4.8.7.6 Perform the following:

(a) Turn the OPERATION FUNCTION switch to TORP PWR OFF.
(b) Connect a second jumper from AC to K of the OUTPUT.
(c) Turn the OPERATION FUNCTION switch to TORP PWR ON.

4.8.7.6.1 The following lamps shall be on: STRATUM LEVEL, HORIZ SEARCH, PORT, STBD, STR, HOMING, COMB, NONSYNC CONTROL, NONSYNC SCAN, SPEED, MEDIUM, TORP PWR ON, BATTERY HEATER, PHASE ABC, READY, AND SYNC SCAN.

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:41Z
Check the source to verify that this is the current version before use.

4.8.7.7 Press DEPTH SET ORDER button and rotate ORDER ADJUST controls from one stop to the other. The OFF lamp shall be on for all numbers except those between 0000.0 and 0999.9, inclusive.

4.8.7.8 Perform the following:

(a) Turn the OPERATION FUNCTION switch to TORP PWR OFF.
(b) Connect a third jumper from AC to f of the OUTPUT.
(c) Turn the OPERATION FUNCTION switch to TORP PWR ON.

4.8.7.8.1 The ANTICAPTURE lamp shall be on.

4.8.7.9 Turn the OPERATION FUNCTION switch to TORP PWR OFF and remove the three jumpers from the OUTPUT.

4.8.8 Mk 5 switchbox test.

4.8.8.1 Turn the OPERATION FUNCTION switch to TORP PWR ON. The voltage from pin A(+) to pin B(-) of the Mk 5 switchbox receptacle shall be 24 ± 2.0 VDC.

4.8.8.2 Turn the OPERATION FUNCTION switch to TORP PWR OFF.

4.8.9 Firing circuit test.

4.8.9.1 Turn the OPERATION FUNCTION switch to TORP PWR ON, operate the FIRE switch, and hold. The following shall be noted:

(a) TORP PWR OFF lamp: on

(b) BATTERY HEATER lamp: on

(c) OUTPUT: 0.1 VDC (max) from AC(+) to AD; 1 VAC (max) between AA and AB, AA and F, and AB and F.

4.8.9.2 Release the FIRE switch.

4.8.9.2.1 The FIRE lamp shall go off and the READY lamp shall come on.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the currentversion before use.

4.8.9.2.2 Turn the OPERATION FUNCTION switch to TORP PWR OFF then to TORP PWR ON.

4.8.9.2.3 The TORP PWR ON and READY lamps shall be on.

4.8.9.3 Perform the following:

(a) Turn the OPERATION FUNCTION switch to SELF TEST.
(b) Operate the FIRE switch and hold.
(c) Rotate the OPERATION FUNCTION switch through all its positions.

4.8.9.3.1 The FIRE lamp shall be on during 4.8.9.3(c).

4.8.9.3.2 Release the FIRE switch.

4.8.9.4 Perform the following:

(a) Turn the OPERATION FUNCTION switch to TORP PWR OFF.
(b) Turn the OPERATION FUNCTION switch to TORP PWR ON.
(c) Set the TUBE SOLENOID circuit breaker OFF.
(d) Operate the FIRE switch and hold.

4.8.9.4.1 The TORP PWR ON lamp shall remain on during 4.8.9.4(d).

4.8.9.4.2 Perform the following:

(a) Release the FIRE switch.
(b) Set the TUBE SOLENOID circuit breaker ON.
(c) Turn the OPERATION FUNCTION switch to TORP PWR OFF.

4.8.9.5 Perform the following:

(a) Connect a 250-ohm rheostat and a 10-amp (full-scale) AC ammeter in series from AK to u of the OUTPUT.

(b) Turn the OPERATION FUNCTION switch to TORP PWR ON.

(c) Operate the FIRE switch and hold.

(d) Adjust rheostat to obtain a reading of 5 amps on the ammeter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.8.9.5.1 The meter reading from AK to u of the OUTPUT shall be from 106 to 121 VAC.

4.8.9.5.2 Release the FIRE switch, turn the OPERATION FUNCTION switch to TORP PWR OFF, and remove the rheostat and ammeter from the OUTPUT.

4.8.10 Tube solenoid test.

4.8.10.1 Perform the following:

(a) Connect a 250-ohm rheostat and a 25-amp (full-scale) AC ammeter in series from pin A to pin B of the TUBE SOLENOID receptacle.

(b) Turn the OPERATION FUNCTION switch to TORP PWR ON.

(c) Operate the FIRE switch and hold.

(d) Adjust the rheostat to obtain a reading of 15 amps on the AC ammeter.

4.8.10.1.1 The meter reading between pins A and B of the TUBE SOLENOID receptable shall be from 106 to 121 VAC.

4.8.10.1.2 Perform the following:

(a) Release the FIRE switch.
(b) Turn the OPERATION FUNCTION switch to TORP PWR OFF.
(c) Remove the rheostat and ammeter from TUBE SOLENOID receptacle.

CAUTION

When performing the following DC power test, do not leave the OPERATION FUNCTION switch in the TORP PWR ON position for more than 5 seconds or the power supply may be destroyed.

4.8.11 DC power test.

4.8.11.1 Perform the following:

(a) Connect a 2.5-ohm 50-watt resistor from AC to AD of the OUTPUT.

(b) Turn the OPERATION FUNCTION switch to TORP PWR ON for no more than 5 seconds.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

4.8.11.2 The meter reading from AC(+) to AD of the OUTPUT shall be 18 VDC (min) to 32 VDC (max).

4.8.11.3 Turn the OPERATION FUNCTION switch to TORP PWR OFF within 5 seconds and remove the resistor from the OUTPUT.

4.8.12 Wire guide test.

4.8.12.1 Connect a jumper from A to B of the WIRE GUIDE receptacle. The resistance between AH and AJ of the OUTPUT shall be 0.5 ohm (max).

4.8.12.2 Remove the jumper on the WIRE GUIDE receptacle. The resistance between AH and AJ of the OUTPUT shall be 1 megohm (min).

4.8.13 Mechanical stops test.

4.8.13.1 Perform the following:

(a) Turn the OPERATION FUNCTION switch to SELF TEST.
(b) Press ENABLE RUN ORDER button.

4.8.13.2 Rotate the ORDER ADJUST COARSE control counterclockwise. The mechanical stop shall occur when the indicator reads between 16600 and 19400.

4.8.13.3 Rotate the ORDER ADJUST COARSE control clockwise. The mechanical stop shall occur when the indicator reads between 22600 and 28400.

4.8.13.4 Turn the OPERATION FUNCTION switch to TORP PWR OFF.

4.8.14 Null indicator test.

4.8.14.1 Perform the following:

(a) Turn the OPERATION FUNCTION switch to SELF TEST.

(b) Press the SELF TEST 1/SELF TEST 2 button to obtain SELF TEST 2.

(c) Press GYRO ANGLE ORDER.

(d) Rotate the ORDER ADJUST control for an indicator reading of 000.00.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

(e) Press SCAN/SET button to obtain SCAN.

(f) Rotate ORDER ADJUST FINE control to obtain a half-scale deflection (either side) of the ORDER ERROR meter.

(g) Press the METER SCALE button.

4.8.14.1.1 The ORDER ERROR meter shall be deflected full scale to the same side as in 4.8.14.1(f). Release the METER SCALE button.

4.8.14.2 Using the SELF TEST 1/SELF TEST 2, ENABLE RUN ORDER, DEPTH SET ORDER, and GYRO ANGLE ORDER buttons, obtain each of the test set conditions listed in the tabulation below. For each condition, use the ORDER ADJUST controls and the ORDER ERROR meter to determine the null position. The null position is defined as the center of the digital indicator range over which the ORDER ERROR meter reads center-scale, or zero error. For each condition, the null position shall be as specified in the tabulation below.

	Condition	Null position
4.8.14.2.1	SELF TEST 2, GYRO ANGLE ORDER	359.95 to 000.05
4.8.14.2.2	SELF TEST 2, GYRO ANGLE ORDER	004.95 to 005.05
4.8.14.2.3	SELF TEST 2, DEPTH SET ORDER	0049.5 to 0050.5
4.8.14.2.4	SELF TEST 2, DEPTH SET ORDER	0199.5 to 0200.5
4.8.14.2.5	SELF TEST 2, ENABLE RUN ORDER	12950. to 13050.
4.8.14.2.6	SELF TEST 2, ENABLE RUN ORDER	30950. to 31050.
4.8.14.2.7	SELF TEST 1, ENABLE RUN ORDER	00950. to 01050.
4.8.14.2.8	SELF TEST 1, ENABLE RUN ORDER	18950. to 19050.
4.8.14.2.9	SELF TEST 1, GYRO ANGLE ORDER	359.50 to 000.50
4.8.14.2.10	SELF TEST 1, GYRO ANGLE ORDER	179.50 to 180.50
4.8.14.2.11	SELF TEST 1, DEPTH SET ORDER	0045.0 to 0055.0
4.8.14.2.12	SELF TEST 1, DEPTH SET ORDER	1545.0 to 1555.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.8.14.3 Turn the OPERATION FUNCTION switch to TORP PWR OFF.

CAUTION

During the following nonsynchronous functional test, voltage is present on the OUTPUT receptacle pins, therefore exercise caution. The steps of this test must be done exactly in sequence. If any switch or jumper is moved out of order, the entire test shall be repeated.

4.8.15 Nonsynchronous functional test.

4.8.15.1 Perform the following:

(a) Connect a jumper from AC to AL to K of the OUTPUT.
(b) Connect a second jumper from AD to U of the OUTPUT.
(c) Set the NONSYNC CONTROL at SET.
(d) TURN THE OPERATION FUNCTION switch to TORP PWR ON.
(e) Remove the OUTPUT jumper from AC to AL to K.

4.8.15.1.1 The following lamps shall be on: SPEED, STRATUM LEVEL, HOMING, HORIZ SEARCH, PORT, STBD, STR, NONSYNC CONTROL, and NONSYNC SCAN.

4.8.15.2 Connect a jumper from AC to G of the OUTPUT momentarily, and remove it. The lamps listed in 4.8.15.1.1 shall remain on except PORT, STBD, and STR.

4.8.15.3 Connect a jumper from AC of the OUTPUT to the other OUTPUT pins listed in the table below, one at a time. During each connection, the specified indicator shall come on. When the connection is broken, the indicator shall go off.

	OUTPUT pin	Indicator	Function
4.8.15.3.1	t	LOW	SPEED
4.8.15.3.2	AL	MEDIUM	SPEED
4.8.15.3.3	z	HIGH	SPEED
4.8.15.3.4	V	NO	STRATUM LEVEL

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

4.8.15.3.5	c	ABOVE	STRATUM LEVEL
4.8.15.3.6	1	BELOW	STRATUM LEVEL
4.8.15.3.7	G	PORT	HORIZ SEARCH
4.8.15.3.8	A	STBD	HORIZ SEARCH
4.8.15.3.9	B	STR	HORIZ SEARCH
4.8.15.3.10	C	LISTEN	HOMING
4.8.15.3.11	D	ECHO	HOMING
4.8.15.3.12	K	COMB	HOMING

4.8.15.3.13 Remove the jumper from the OUTPUT.

4.8.15.4 Turn the NONSYNC CONTROL to SCAN and back to SET. The SET lamp shall now be on.

4.8.15.5 Perform the following:

(a) Set SPEED at LOW.
(b) Set STRATUM LEVEL at NO.
(c) Set HORIZ SEARCH at PORT.
(d) Set HOMING at LISTEN.

4.8.15.6 The following shall be noted:

(a) OUTPUT: 24 ± 2.0 VDC from AC(+) to each of the following pins (-): t, V, G, and C

(b) OUTPUT: 27 VDC (min) to 32 VDC (max) from pin AC(+) to each of the following pins (-): AL, z, c, 1, A, B, D, K, and E.

4.8.15.7 Perform the following:

(a) Set SPEED at MEDIUM.
(b) Set STRATUM LEVEL at ABOVE.
(c) Set HORIZ SEARCH at STBD.
(d) Set HOMING at ECHO.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.8.15.8 The following shall be noted:

(a) OUTPUT: 24 ± 2.0 VDC from AC(+) to each of the following pins (-): AL, c, A, and D

(b) OUTPUT: 27 VDC (min) to 32 VDC (max) from pin AC(+) to each of the following pins (-): t, z, V, l, G, B, C, K, and E.

4.8.15.9 Perform the following:

(a) Set SPEED at HIGH.
(b) Set STRATUM LEVEL at BELOW.
(c) Set HORIZ SEARCH at STR.
(d) Set HOMING at COMB.

4.8.15.10 The following shall be noted:

(a) OUTPUT: 24 ± 2.0 VDC from AC(+) to each of the following pins (-): z, l, B, and K

(b) OUTPUT: 27 VDC (min) to 32 VDC (max) from AC(+) to each of the following pins (-): t, AL, V, c, G, A, C, D, and E.

4.8.15.11 Perform the following:

(a) Turn ANTICAPTURE switch to position 1.
(b) Set NONSYNC CONTROL to SCAN.
(c) Connect a jumper from pin AC to f of the OUTPUT.

4.8.15.11.1 The ANTICAPTURE indicator lamp must come on.

4.8.15.12 Using the decade resistor of 4.3(g), obtain each of the resistances specified below, one at a time between OUTPUT pin f and pin o. For each resistance, turn the ANTICAPTURE switch through all its positions. When the ANTICAPTURE switch is at the position corresponding to the attached resistance as specified below, all four ANTICAPTURE indicator quadrants shall come on.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

Resistor (ohms)	Position
1.00 k, 1%	1
1.40 k, 1%	2
1.96 k, 1%	3
2.74 k, 1%	4
3.83 k, 1%	5
5.36 k, 1%	6
7.50 k, 1%	7
715, 1%	8

4.8.15.13 Turn the NONSYNC CONTROL to SET. Again connect each of the eight resistances specified in 4.8.15.12 from f to o of the OUTPUT. With each resistance connected, rotate the ANTICAPTURE switch through its eight positions.

4.8.15.13.1 The voltage from AC(+) to v(-) of the OUTPUT shall be 24 ± 2.0 VDC when the switch position and resistance connected correspond according to the table of 4.8.15.12.

4.8.15.13.2 The voltage from AC(+) to v(-) of the OUTPUT shall be 0.1 VDC (max) in all positions except the corresponding ones.

4.8.15.14 Turn the OPERATION FUNCTION switch to TORP PWR OFF and remove all jumpers and the decade resistor from the OUTPUT.

4.8.16 Interlock test.

4.8.16.1 Perform the following:

(a) Connect a jumper from AC to AL of the OUTPUT.
(b) Connect a second jumper from AD to U of the OUTPUT.
(c) Press GYRO ANGLE ORDER BUTTON.
(d) Set SPEED at MEDIUM.
(e) Turn the OPERATION FUNCTION switch to TORP PWR ON.

4.8.16.1.1 The SYNC SCAN and NONSYNC SCAN lamps shall be on.

4.8.16.2 Perform the following:

(a) Press SYNC SCAN/SET button to obtain SET condition.
(b) Set NONSYNC CONTROL to obtain SET condition.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

(c) Turn the OPERATION FUNCTION switch to TORP PWR OFF.
(d) Turn the OPERATION FUNCTION switch to TORP PWR ON.

4.8.16.2.1 The SYNC SCAN and NONSYNC SCAN lamps shall be on.

4.8.16.3 Perform the following:

(a) Press the SCAN/SET button to obtain SET condition.
(b) Press ENABLE RUN ORDER button.

4.8.16.3.1 The SYNC SCAN and NONSYNC SCAN lamps shall be on.

4.8.16.4 Press the SYNC SCAN/SET button twice. The SYNC SET lamp shall be on.

4.8.16.5 Turn the OPERATION FUNCTION switch to TORP PWR OFF and remove the jumpers from the OUTPUT.

4.8.17 Bias transformer test.

4.8.17.1 Perform the following:

(a) Open access door on top of test set.

(b) Turn the OPERATION FUNCTION switch to SELF TEST.

(c) Press SELF TEST 1/SELF TEST 2 button to obtain SELF TEST 1.

(d) Press GYRO ANGLE ORDER button.

(e) Turn ORDER ADJUST CONTROLS to obtain a null on the ORDER ERROR meter.

4.8.17.2 The following shall be noted:

(a) 0.5 VAC (max) from GRD (black) to L. S. (red) test points on the GYRO function amplifier

(b) 2.5 ± 0.5 VAC from GRD (black) to L. S. BIASED (yellow) test points on the GYRO function amplifier.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

4.8.17.3 Turn the OPERATION FUNCTION switch to TORP PWR OFF.

4.8.18 Amplifier gain tests.

4.8.18.1 Perform the following:

(a) Connect a jumper from AD to O of the OUTPUT.

(b) Connect a second jumper from AB to T to P of the OUTPUT.

(c) Connect the decade inductor, set at 225 mH, from AA to H of the OUTPUT.

(d) Connect a 960-ohm, 25-watt resistor from L to M of the OUTPUT.

(e) Connect a jumper from COMMON to GYRO LOW SPEED test points on the front panel of the test set.

(f) Press GYRO ANGLE ORDER button.

(g) Turn the OPERATION FUNCTION switch to TORP PWR ON.

(h) Press the SYNC SCAN/SET button to obtain SET condition.

(i) Rotate the ORDER ADJUST controls to change the input voltage as required.

4.8.18.1.1 The gain of the two-speed gyro loop shall be as specified below. It shall be measured twice, first with the ORDER ERROR meter deflected to the left, and second with the meter deflected to the right. The input voltage is measured from GYRO HIGH SPEED to COMMON test points on the front panel. The output voltage is measured from L to M (across the 960-ohm resistor) of the OUTPUT.

	Output (VAC)	
Input (VAC)	Min	Max
2	14	40
5	50	85
10	70	130

4.8.18.1.2 Turn the OPERATION FUNCTION switch to TORP PWR OFF, disconnect all jumpers except the one from AD to O on the OUTPUT, and disconnect the inductor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.8.18.1.3 Perform the following:

(a) Connect a jumper from AA to I to T of the OUTPUT.

(b) Connect the 225-mH inductor from F to J of the OUTPUT.

(c) Connect a jumper from COMMON to GYRO HIGH SPEED test points on the front panel.

(d) Turn the OPERATION FUNCTION switch to TORP PWR ON.

(e) Press the SYNC SCAN/SET button to obtain SET condition.

4.8.18.1.4 The gain of the two-speed gyro loop shall be as specified below. It shall be measured twice, first with the ORDER ERROR meter deflected to the left, and second with the meter deflected to the right. The input voltage is measured from the GYRO LOW SPEED to COMMON test points on the front panel. The output voltage is measured from L to M (across the 960-ohm resistor) of the OUTPUT.

Input (VAC)	Output (VAC) Min	Max
4	50	110
10	75	130

4.8.18.1.5 Turn the OPERATION FUNCTION switch to TORP PWR OFF. Remove all connections except the jumper from AD to O of the OUTPUT.

4.8.18.2 Perform the following:

(a) Connect a second jumper from AB to T to r of the OUTPUT.

(b) Connect the 225-mH inductor from AA to x of the OUTPUT.

(c) Connect a 960-ohm, 25-watt resistor from L to d of the OUTPUT.

(d) Press the DEPTH SET ORDER button.

(e) Turn the OPERATION FUNCTION switch to TORP PWR ON.

(f) Press the SYNC SCAN/SET button to obtain SET condition.

(g) Rotate the ORDER ADJUST controls to change the input voltage as required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.8.18.2.1 The gain of the high-speed depth loop shall be as specified below. It shall be measured twice, first with the ORDER ERROR meter deflected to the left, and second with the meter deflected to the right. The input voltage is measured from DEPTH HIGH SPEED and COMMON test points on the front panel. The output voltage is measured from L to d (across the 960-ohm resistor) of the OUTPUT.

	Output (VAC)	
Input (VAC)	Min	Max
1	10	35
2	40	70
4	70	130

4.8.18.2.2 Turn the OPERATION FUNCTION switch to TORP PWR OFF and remove all connections except the jumper from AD to O of the OUTPUT.

4.8.18.3 Perform the following;

(a) Connect a second jumper from AB to q to T of the OUTPUT.

(b) Connect the 225-mH inductor from AA to p of the OUTPUT.

(c) Connect the 960-ohm, 25-watt resistor from L to m of the OUTPUT.

(d) Press the ENABLE RUN ORDER button.

(e) Turn the OPERATION FUNCTION switch to TORP PWR ON.

(f) Press the SYNC SCAN/SET button to obtain SET condition.

(g) Use the ORDER ADJUST controls to vary the input voltage as required.

4.8.18.3.1 The gain of the enable loop shall be as specified below. It shall be measured twice, first with the ORDER ERROR meter deflected to the left, and second with the meter deflected to the right. The input voltage is measured from the GRD (black) to L. S. SIGNAL (red) test points on the DEPTH function amplifier. The output voltage is measured from L to m (across the 960-ohm resistor) of the OUTPUT.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

Input (VAC)	Output (VAC) Min	Max
0.15	25	55
0.30	55	83
0.40	70	130

4.8.18.3.2 Turn the OPERATION FUNCTION switch to TORP PWR OFF and remove all connections.

4.8.18.4 Perform the following:

(a) Connect a jumper from AD to U to Y of the OUTPUT.

(b) Connect a second jumper from AA to I to T of the OUTPUT.

(c) Connect the 225-mH inductor from F to J of the OUTPUT.

(d) Connect a 960-ohm, 25-watt resistor from L to M of the OUTPUT.

(e) Press the GYRO ANGLE ORDER button.

(f) Turn the OPERATION FUNCTION switch to TORP PWR ON.

(g) Press the SYNC SCAN/SET button to obtain SET condition.

(h) Use the ORDER ADJUST controls to vary the input voltage as required.

4.8.18.4.1 The gain of the single-speed gyro loop shall be as specified below. It shall be measured twice, first with the ORDER ERROR meter deflected to the left, and second with the meter deflected to the right. The input voltage is measured from the GRD (black) to L. S. (red) test points on the GYRO function amplifier. The output voltage is measured from L to M (across the 960-ohm resistor) of the OUTPUT.

Input (VAC)	Output (VAC) Min	Max
0.5	35	63
1	65	100
2	70	130

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.8.18.4.2 Turn the OPERATION FUNCTION switch to TORP PWR OFF and remove all connections except the jumper from AD to U to Y of the OUTPUT.

4.8.18.5 Perform the following:

(a) Connect a second jumper from AA to T to b of the OUTPUT.

(b) Connect the 225-mH inductor from F to s of the OUTPUT.

(c) Connect a 960-ohm, 25-watt resistor across the DEPTH AMP OUT test points on the front panel.

(d) Press the DEPTH SET ORDER button.

(e) Turn the OPERATION FUNCTION switch to TORP PWR ON.

(f) Press the SYNC SCAN/SET button to obtain SET condition.

(g) Use the ORDER ADJUST controls to vary the input voltage as required.

4.8.18.5.1 The low-speed depth loop gain shall be as specified below. It shall be measured twice, first with the ORDER ERROR meter deflected to the left, and second with the meter deflected to the right. The input voltage is measured from the GRD (black) to the L. S. SIGNAL (red) test points on the DEPTH function amplifier. The OUTPUT voltage is measured across the 960-ohm resistor.

	Output (VAC)	
Input (VAC)	Min	Max
0.15	25	55
0.30	55	83
0.40	70	130

4.8.18.5.2 Turn the OPERATION FUNCTION switch to TORP PWR OFF and remove all connections.

4.8.19 Phase and amplitude test.

4.8.19.1 Perform the following:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

(a) Disconnect the 400-Hz power and prepare to connect the variable-voltage 400-Hz supply.

(b) Set the OPERATION FUNCTION switch to TORP PWR ON.

4.8.19.1.1 The LOW lamp of the 3-PHASE VOLTAGE indicator shall be on and the TORP PWR OFF LAMP shall be on.

4.8.19.2 Perform the following:

(a) Turn the OPERATION FUNCTION switch to TORP PWR OFF.
(b) Reduce 400-Hz voltage to 80 volts or less.
(c) Connect reduced voltage to POWER INPUT.
(d) Turn the OPERATION FUNCTION switch to TORP PWR ON.

4.8.19.2.1 The following shall be noted:

(a) TORP PWR OFF lamp: on

(b) OUTPUT: 1 VAC (max) from AA to AB, AA to F, and AB to F of the OUTPUT.

4.8.19.3 Increase the 400-Hz voltage gradually. At 95 ± 5 VAC, the LOW lamp of the 3-PHASE VOLTAGE indicator shall go off and the TORP PWR ON and PHASE ABC lamps shall come on.

4.8.19.4 Increase the 400-Hz voltage further. At 135 ± 5 VAC, the TORP PWR ON and PHASE ABC lamps shall go off and the TORP PWR OFF and HIGH lamps shall come on. At this point, there shall be 1.0 VAC (max) from AA to AB, AA to F, and AB to F of the OUTPUT.

4.8.19.5 Turn the 400-Hz voltage supply off. The HIGH lamp of the 3-PHASE VOLTAGE indicator shall go off and the PHASE CBA lamp shall come on.

4.8.19.6 Operate the AC POWER circuit breaker OFF and ON. The PHASE CBA lamp shall go off and the LOW lamp of the 3-PHASE VOLTAGE indicator shall come on. Turn the OPERATION FUNCTION switch to TORP PWR OFF.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.8.19.7 Disconnect the variable-voltage 400-Hz supply and connect the 115-volt, 400-Hz supply with any two of the three leads reversed. Turn the OPERATION FUNCTION switch to TORP PWR ON. The PHASE CBA indicator and the TORP PWR OFF lamp shall come on. There shall be 1 VAC (max) between AA and AB, AA and F, and AB and F of the OUTPUT. Set the OPERATION FUNCTION switch to TORP PWR OFF and reconnect the 400-Hz power as specified in 3.4.1.

4.8.20 Synchronous off test.

4.8.20.1 Perform the following:

(a) Connect a jumper from AD to O of the OUTPUT.

(b) Connect a second jumper from P to T of the OUTPUT.

(c) Connect a third jumper from AB to H of the OUTPUT.

(d) Press GYRO ANGLE ORDER button.

(e) Turn the OPERATION FUNCTION switch to SELF TEST.

(f) Turn ORDER ADJUST controls to obtain 159.00 on the GYRO ANGLE ORDER indicator.

(g) Turn the OPERATION FUNCTION switch to TORP PWR ON.

4.8.20.1.1 The following shall be noted:

(a) ORDER ERROR METER: shows deflection

(b) SYNC SCAN lamp: on

(c) 1 VAC (max) across the OUTPUT (white) test points on the GYRO function amplifier.

4.8.20.1.2 Press the SYNC SCAN/SET button to obtain a SET condition. The ORDER ERROR meter shall show a deflection and the voltage across the OUTPUT (white) test points of the GYRO function amplifier shall be 75 VAC (min) to 115 VAC (max).

4.8.20.1.3 Adjust the ORDER ADJUST CONTROLS to obtain a reading of 161.00 on the GYRO ANGLE ORDER indicator. The following shall be noted:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

(a) ORDER ERROR meter: deflected

(b) SYNC SET lamp: on

(c) SYNC CONTROL OFF lamp: on

(d) 1 VAC (max) across the OUTPUT (white) test points on the GYRO function amplifier.

4.8.20.1.4 Turn the ORDER ADJUST controls to obtain a reading of 159.00 on the GYRO ANGLE ORDER indicator. The voltage across the OUTPUT (white) test points on the GYRO function amplifier shall be 75 VAC (min) to 115 VAC (max).

4.8.20.1.5 Press the SYNC OFF button. The following shall be noted:

(a) SYNC OFF lamp: on

(b) SYNC SET lamp: on

(c) ORDER ERROR meter: not deflected

(d) 1 VAC (max) between the OUTPUT (white) test points on the GYRO function amplifier.

4.8.20.1.6 Rotate the ORDER ADJUST COARSE control through at least two complete revolutions. The GYRO ANGLE ORDER indicator shall still read 159.00.

4.8.20.2 Turn the OPERATION FUNCTION switch to TORP PWR OFF and remove all connections.

4.8.21 Relay K20 operation test.

4.8.21.1 Perform the following:

(a) Turn the OPERATION FUNCTION switch to SELF TEST.

(b) Press the DEPTH SET ORDER button.

(c) Press SELF TEST 1/SELF TEST 2 button to obtain SELF TEST 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

(d) Turn ORDER ADJUST controls to obtain a deflection on the ORDER ERROR meter.

(e) Connect an isolated power supply set at 24 VDC from AC(+) to Y(-) on the OUTPUT.

4.8.21.1.1 The ORDER ERROR meter shall now be deflected to the opposite side from that in 4.8.21.1(d).

4.8.21.2 Turn the OPERATION FUNCTION switch to TORP PWR OFF and remove all connections.

4.8.22 Relay K21 operation test.

4.8.22.1 Perform the following:

(a) Turn the OPERATION FUNCTION switch to SELF TEST.

(b) Press the GYRO ANGLE ORDER button.

(c) Press the SELF TEST 1/SELF TEST 2 button to obtain SELF TEST 1.

(d) Connect an isolated power supply set at 24 VDC from AC(+) to U(-) of the OUTPUT.

(e) Turn the ORDER ADJUST controls to obtain a deflection on the ORDER ERROR meter.

(f) Connect a jumper from AC to AL of the OUTPUT.

4.8.22.1.1 The ORDER ERROR meter shall deflect to the opposite side from that in 4.8.22.1(e).

4.8.22.2 Turn the OPERATION FUNCTION switch to TORP PWR OFF and remove all connections.

4.8.23 Relay K18 operation test.

4.8.23.1 Perform the following:

(a) Turn the OPERATION FUNCTION switch to SELF TEST.

(b) Press the DEPTH SET ORDER button.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

(c) Press the SELF TEST 1/SELF TEST 2 button to obtain SELF TEST 1.

(d) Turn the ORDER ADJUST controls to obtain a deflection on the ORDER ERROR meter.

(e) Connect an isolated power supply set at 24 VDC from AC(+) to U(-) of the OUTPUT.

4.8.23.1.1 The ORDER ERROR meter shall now be deflected to the opposite side from that in 4.8.23.1(d).

4.8.23.2 Turn the OPERATION FUNCTION switch to TORP PWR OFF and remove all connections.

4.9 Accuracy inspection procedure.

4.9.1 Single-speed gyro.

4.9.1.1 Perform the following:

(a) Connect a jumper from AD to U of the OUTPUT.

(b) Connect the ratio transformer of 4.3(c) as follows:

S_1 to I of the OUTPUT
S_2 to J of the OUTPUT
S_3 to T of the OUTPUT
R_1 to AA of the OUTPUT
R_2 to F of the OUTPUT.

(c) Turn the OPERATION FUNCTION switch to TORP PWR ON.

(d) Press the SCAN/SET button to obtain a SCAN condition.

(e) Press the GYRO ANGLE ORDER button.

4.9.1.2 Perform the following for each setting of the ratio transformer in the tabulation below:

(a) Set the ratio transformer to the listed angle.

(b) Turn the ORDER ADJUST controls to the corresponding "true position" on the digital indicators.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

(c) Press the METER SCALE button and hold.

(d) Using the ORDER ADJUST FINE control, note the range of digital indicator readings over which the ORDER ERROR meter reads zero error.

(e) Release the METER SCALE button.

(f) Record the central indicator reading of the range noted in step (d) as "indicated position," for that angle.

Transformer setting	True position	Indicated position
0°	000.00	
30°	030.00	
60°	060.00	
90°	090.00	
120°	120.00	
150°	150.00	
180°	180.00	
210°	210.00	
240°	240.00	
270°	270.00	
300°	300.00	
330°	330.00	

4.9.1.3 The difference between true and indicated readings as noted above shall be no more than 000.25 for any one reading. The arithmetic mean of the 12 absolute differences shall be no more than 000.10.

4.9.1.4 Turn the OPERATION FUNCTION switch to TORP PWR OFF and remove the ratio transformer connections. Do not remove the jumper.

4.9.2 Low-speed depth.

4.9.2.1 Perform the following:

(a) Connect the ratio transformer of 4.3(c) as follows:

S_1 to b of the OUTPUT
S_2 to $\overline{s}$ of the OUTPUT
S_3 to $\overline{T}$ of the OUTPUT
R_1 to AA of the OUTPUT
R_2 to F of the OUTPUT.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:41Z
Check the source to verify that this is the current version before use.

(b) Turn the OPERATION FUNCTION switch to TORP PWR ON.

(c) Press the DEPTH SET ORDER button.

4.9.2.2 For each of the ratio transformer settings listed in the tabulation below, follow the procedure of 4.9.1.2 (a) through (f) to obtain the "indicated" positions for each angle.

Transformer setting	True position	Indicated position
0°	0050.0	
30°	0300.0	
60°	0550.0	
90°	0800.0	
120°	1050.0	
150°	1300.0	
180°	1550.0	
210°	1800.0	
240°	2050.0	
270°	2900.0	
300°	3150.0	
330°	3400.0	

4.9.2.3 The difference between true and indicated readings as noted above shall be no more than 0002.5 for any one reading. The arithmetic mean of the 12 absolute differences shall be no more than 0001.0.

4.9.2.4 Turn the OPERATION FUNCTION switch to TORP PWR OFF and remove the jumper and ratio transformer.

4.9.3 Enable run order.

4.9.3.1 Perform the following:

(a) Connect the ratio transformer of 4.3(c) as follows:

S_1 to $\underline{q}$ of the OUTPUT
S_2 to $\underline{p}$ of the OUTPUT
S_3 to $\overline{T}$ of the OUTPUT
R_1 to AA of the OUTPUT
R_2 to AB of the OUTPUT

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

(b) Turn the OPERATION FUNCTION switch to TORP PWR ON.

(c) Press the ENABLE RUN ORDER button.

4.9.3.2 For each of the ratio transformer settings listed in the tabulation below, follow the procedure of 4.9.1.2 (a) through (f) to obtain the "indicated" positions for each angle.

Transformer setting	True position	Indicated position
0°	01000.	
30°	04000.	
60°	07000.	
90°	10000.	
120°	13000.	
150°	16000.	
180°	19000.	
210°	22000.	
240°	25000.	
270°	28000.	
300°	31000.	
330°	34000.	

4.9.3.3 The difference between true and indicated readings as noted above shall be no more than 00025. for any one reading. The arithmetic mean of the 12 absolute differences shall be no more than 00010.

4.9.3.4 Turn the OPERATION FUNCTION to TORP PWR OFF and remove the ratio transformer.

4.9.4 High-speed depth.

4.9.4.1 Perform the following:

(a) Connect a jumper from AD to O of the OUTPUT.

(b) Connect the ratio transformer of 4.3(c) as follows:

S_1 to $\underline{r}$ of the OUTPUT
S_2 to $\underline{x}$ of the OUTPUT
S_3 to $\overline{T}$ of the OUTPUT
R_1 to AA of the OUTPUT
R_2 to AB of the OUTPUT.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

(c) Turn the OPERATION FUNCTION switch to TORP PWR ON.

(d) Press the DEPTH SET ORDER button.

4.9.4.2 For each of the ratio transformer settings listed in the tabulation below, follow the procedure of 4.9.1.2 (a) through (f) to obtain the "indicated" positions for each angle.

Transformer setting	True position	Indicated position
0°	0050.0	
30°	0075.0	
60°	0100.0	
90°	0125.0	
120°	0150.0	
150°	0175.0	
180°	0200.0	
210°	0225.0	
240°	0250.0	
270°	0275.0	
300°	0000.0	
330°	0025.0	

4.9.4.3 The difference between true and indicated readings as noted above shall be no more than 0000.2 for any one reading. The arithmetic mean of the 12 absolute differences shall be no more than 0000.1.

4.9.4.4 Turn the OPERATION FUNCTION switch to TORP PWR OFF and remove the transformer. Do not remove the jumper.

4.9.5 *Two-speed gyro.*

4.9.5.1 Perform the following:

(a) Connect the ratio transformer of 4.3(c) as follows:

S_1 to P of the OUTPUT
S_2 to H of the OUTPUT
S_3 to T of the OUTPUT
R_1 to AA of the OUTPUT
R_2 to AB of the OUTPUT

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

(b) Turn the OPERATION FUNCTION switch to TORP PWR ON.

(c) Press the GYRO ANGLE ORDER button.

4.9.5.2 For each of the ratio transformer settings in the tabulation below, follow the procedure of 4.9.1.2 (a) through (f) to obtain the "indicated" positions for each angle.

Transformer setting	True position	Indicated position
0°	000.00	
30°	000.83	
60°	001.67	
90°	002.50	
120°	003.33	
150°	004.17	
180°	005.00	
210°	005.83	
240°	006.67	
270°	007.50	
300°	008.33	
330°	009.17	

4.9.5.3 The difference between true and indicated readings as noted above shall be no more than 000.02 for any one reading. The arithmetic mean of the 12 absolute readings shall be no more than 000.01.

4.9.5.4 Turn the OPERATION FUNCTION switch to TORP PWR OFF and remove the jumper and ratio transformer.

4.10 Environmental test procedures.

4.10.1 Vibration.

4.10.1.1 Shipping and handling. The test set, in a nonoperating condition and stowed in its case, shall be subjected to the vibration test specified in MIL-STD-810, method 514, procedure X, curve AB from 5 to 52 Hz, with sweep time as shown in table 514-II, schedule IV.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

4.10.1.2 Shipboard. The test set shall be subjected to the shipboard vibration test specified in MIL-STD-167, using resonant frequencies (if any) found in the test procedure specified. The test set shall be operating during this test, and shall be subjected to the performance test procedure of 4.8, herein, while vibrating in each of its three principal axes.

4.10.2 Shock. The test set shall be subjected to the shock test specified in MIL-STD-810, method 516, procedure I, wave shape 516-2, amplitude (a), duration (c).

4.10.3 Temperature.

4.10.3.1 Low temperature. The test set, less packaging or packing, shall be placed in a test chamber maintained at -65° ± 3° Fahrenheit for a period of 24 ± 2 hours. At the end of that time, the temperature shall be increased to 32° ± 3° Fahrenheit over a period of 3 ± 0.5 hours. The test set shall be held at 32° ± 3° Fahrenheit for 2 ± 0.5 hours, and then shall be subjected to the performance test procedure of 4.8, herein, while at 32° ± 3° Fahrenheit.

4.10.3.2 High temperature. The test set, less packaging or packing, shall be subjected to 160° ± 3° Fahrenheit for 24 ± 2 hours. At the end of that time, the test set shall be placed in an ambient environment and left for not less than 4 hours before being subjected to further tests.

4.10.4 Humidity.

4.10.4.1 High humidity. The test set, less packaging or packing, shall be subjected to the humidity test specified in MIL-E-5272, procedure I, except the maximum temperature shall be 130° ± 3° Fahrenheit.

4.10.4.2 Low humidity. The test set, less packaging or packing, shall be subjected to the humidity test specified in MIL-E-5272, procedure I, except the maximum temperature shall be 130° ± 3° Fahrenheit, and during the 16-hour cooling-off period of each cycle, the relative humidity shall be reduced to 25% during the first 4 ± 0.4 hours, and maintained at 25% or less for the remaining 12 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.10.4.3 Operation. If specified by the procuring activity, the test set shall be subjected to the performance test procedure of 4.8, herein, while at 130° ± 3° Fahrenheit and in 95% ± 3% relative humidity during the last cycle of the humidity test specified in 4.10.4.1.

4.10.5 Salt-fog. The test set, less packaging or packing, shall be subjected to the salt-fog test specified in MIL-STD-810, method 509, procedure I.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A, unit packaging. The test set shall be packaged in accordance with MIL-P-116, method IIb.

5.1.2 Level C, unit packaging. The test set shall be packaged in accordance with approved commercial practice.

5.2 Packing.

5.2.1 Level A. The test set, packaged as previously specified, shall be packed in boxes conforming to PPP-B-591, overseas type; PPP-B-621, class 2; PPP-B-585, class 3; or PPP-B-601, overseas type.

5.2.2 Level B. The test set, packaged as previously specified, shall be packed in boxes conforming to PPP-B-636, class 1; PPP-B-591, domestic type; PPP-B-621, class 1; PPP-B-585, class 1; or PPP-B-601, domestic type.

5.2.3 Level C. The test set, packaged as previously specified, shall be packed to insure acceptance and safe delivery by common or other carriers for safe transportation, at the lowest rate, to the point of delivery called for in the purchase order.

5.3 Marking. Unless otherwise specified in the contract or purchase order, marking of all containers shall be in accordance with MIL-STD-129. Each container shall be marked with the serial number of the unit it contains.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:41Z
Check the source to verify that this is the current version before use.

5.4 Packaging and packing for storage. Unless otherwise specified in the contract or purchase order, the test set shall be packaged for storage as specified in 5.1.1, and packed for storage as specified in 5.2.1.

6. NOTES

6.1 Intended use. Torpedo Test Set Mk 281 Mod 1 is used to set the synchronous and nonsynchronous functions of the following torpedoes: Mk 14 Mod 5, Mk 16 Mod 8, Mk 37 Mod 0, Mk 37 Mod 2, Mk 37 Mod 3, Mk 45 Mod 1, Mk 45 Mod 2; and Target Torpedo Mk 27 Mod 0.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification, exceptions to this document, applicable drawings, and other pertinent documents.

6.3 Manufacture by Government activities. When test sets are to be manufactured in accordance with this specification by Government activities, the requirements given herein for contractors shall apply to the Government activity.

Custodian:
Navy - OS

Preparing activity:
Navy - OS
(Project Order 1355-N010)

☆U.S. GOVERNMENT PRINTING OFFICE:1971-714-157/12395

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.